

CHARGE DESERT

EV Infrastructure Intelligence Platform
for Identifying **Charging Gaps** and
Prioritizing Potential **Expansion Opportunities**

FROM
CHARGING DESERTS

TO
CONNECTED INDIA

Data-Driven Decisions for a Sustainable EV Future

- IDENTIFY CHARGING GAPS
- PRIORITIZE HIGH-IMPACT LOCATIONS
- DATA-BACKED DECISION MAKING
- ACCELERATE INDIA'S EV REVOLUTION



KEY INSIGHTS (PAN INDIA)

 35K+ EV Registrations (2025)	 15K+ Charging Stations (All Types)	 1.2M+ People Lack Adequate Charging Access	 5 Highway Corridors Analyzed	 30+ High-Potential Locations Identified
--	--	--	--	---



PREPARED BY
KOUSHIK GARG



DOMAIN
DATA ANALYTICS
BUSINESS INTELLIGENCE
GIS • DECISION INTELLIGENCE



APPLICATION
STREAMLIT-BASED
INTERACTIVE ANALYTICAL
PLATFORM

**EXPLORE.
ANALYZE.
TRANSFORM.**

IMPORTANT DATA & MODEL DISCLAIMER

This project is an analytical and portfolio demonstration. The dataset underlying this report is simulated / synthetic data assembled for the ChargeDesert analytical platform.

Accordingly, the findings, scores, rankings and recommendations presented in this report should not be interpreted as official government statistics, verified infrastructure-investment recommendations, or commercially validated site-selection decisions. Metrics calculated directly from source attributes are presented as analytical indicators. Composite scores and priority classifications are illustrative model outputs based on weighted scoring assumptions defined in the project methodology; they require further validation before any real-world deployment decision. This report classifies EV registrations and station/POI/substation attributes as real, observed source fields; arithmetic ratios such as EV penetration, availability rate and EV-per-operational-charger as derived metrics computed from those fields; and demand score, gap score, priority score, priority class, corridor priority score and all narrative recommendations as model outputs.

Executive Summary

India's transition toward electric mobility creates an increasing requirement for accessible, reliable and appropriately distributed charging infrastructure. A simple count of charging stations does not, by itself, indicate whether infrastructure is adequately aligned with EV demand.

ChargeDesert is an EV infrastructure intelligence platform built to analyze the relationship between EV demand, charging infrastructure, geographic accessibility, highway conditions, commercial opportunity and grid readiness, using a combined dataset of EV registrations, charging-station attributes, highway corridor telemetry, points of interest (POI) and power-substation records.

1,32,64,298 EV Registrations (2025)	15,000 Charging Stations (Total)	74.5% Station Availability	1,187 EV per Operational Charger
---	--	--------------------------------------	--

6 States Covered	30 Districts Analyzed	5 Highway Corridors	6 High/Critical Priority Locations
----------------------------	---------------------------------	-------------------------------	--

The analytical layer built from the source data comprises 6 state records, 30 district records, 5 highway corridors (10,000 road segments), 30 location-opportunity records, 60,000 points of interest and 10,000 power substations.

The reference year for EV demand analysis is 2025, selected because it is the most recent complete year in the source data. The source data extends into 2026 (months Apr, Aug, Dec, Feb, Jan, Jul, Jun, Mar, May, Nov, Oct, Sep); these incomplete 2026 observations are excluded from year-over-year growth analysis throughout this report to avoid distortion.

The platform does not treat any single metric as sufficient for infrastructure prioritization. Instead, it combines EV demand, charging supply, infrastructure pressure, charging availability, highway accessibility, freight movement, charging-desert segments, commercial activity and grid readiness into a structured, weighted decision-support framework.

Under the current model, 6 of 30 analyzed districts are classified High priority, 24 Moderate, and 0 Low; no district currently reaches the Critical threshold (≥ 75). Bengaluru Urban (Karnataka), Mumbai and Pune (Maharashtra) rank highest by priority score, and NH-44 ranks highest among the five highway corridors by corridor priority score. These rankings are explicitly model outputs — see Section 29 and the disclaimer above — and should be read as a screening shortlist for further investigation, not as validated investment decisions.

The final deliverable is an interactive Streamlit application (Section 37), supported by this analytical report.

Introduction

Electric vehicles are transforming the transportation ecosystem. As EV adoption increases, charging infrastructure becomes an important component of the broader mobility ecosystem. Infrastructure planning, however, cannot be reduced to answering **"Where are charging stations located?"**

A more useful question is: **"Where does charging infrastructure appear insufficient relative to EV demand, accessibility requirements and infrastructure readiness?"**

This distinction forms the foundation of ChargeDesert.

A region can have charging stations but still experience infrastructure pressure if EV demand is high, operational stations are limited, charger accessibility is poor, highway traffic is significant, charging capacity is insufficient, or commercial and grid opportunities remain underserved. The project therefore treats charging infrastructure as a spatial and business-intelligence problem, rather than simply a station-location problem.

Business Context

The charging ecosystem involves multiple stakeholders with different objectives.

Government and Policymakers

Need to determine where infrastructure gaps exist, which corridors require attention, which regions may require public support, and how infrastructure expansion can be prioritized.

Charging Operators

Need to identify potential expansion markets, underserved areas, high-demand corridors, and commercially attractive locations.

Infrastructure Investors

Need analytical indicators that help evaluate market demand, infrastructure pressure, accessibility, commercial potential, and infrastructure readiness.

Urban and Transport Planners

Need to understand geographic distribution, demand concentration, accessibility, highway coverage, and future infrastructure requirements.

Problem Statement

Existing charging-location platforms primarily answer: **"Where is the nearest charging station?"** ChargeDesert addresses a different question: **"Where may charging infrastructure be insufficient relative to EV demand, accessibility and infrastructure-readiness conditions, and where should potential expansion opportunities be investigated first?"**

The challenge consists of four analytical components:

- Measure EV demand.
- Measure existing charging infrastructure.
- Identify demand–supply and accessibility gaps.
- Prioritize potential opportunities using a transparent decision framework.

Why This Project?

A simple station-count approach has several limitations that motivated this project.

Limitation 1 — Station Count Does Not Represent Demand

A region with many charging stations may still experience pressure if EV demand is extremely high. In this dataset, the national EV-per-operational-charger ratio is 1,187, but this ratio varies substantially across states and districts (Chapter 18–19).

Limitation 2 — Geographic Distribution Matters

Two regions can have identical station counts while having very different accessibility profiles, as shown by the district-level pressure analysis in Chapter 18.

Limitation 3 — Highway Infrastructure Requires Separate Analysis

A charging station located away from a major transport corridor may not adequately serve long-distance traffic. The highway corridor dataset lacks a state/district key and is therefore analyzed independently (Chapter 11).

Limitation 4 — Commercial Feasibility Matters

Potential locations near commercial activity, parking and other facilities may offer different opportunities, captured through the Commercial Activity Score component of the priority model (Chapter 29).

Limitation 5 — Grid Readiness Matters

A potentially attractive location must also be considered in relation to electrical infrastructure — substation proximity, spare capacity and grid reliability (Chapter 27).

Charging infrastructure planning therefore requires multi-dimensional analysis rather than a single station-count metric.

Project Objectives

Primary Objective

Develop an analytical platform for identifying potential EV charging infrastructure gaps and prioritizing locations and corridors for further investigation.

Secondary Objectives

- Analyze EV demand.
- Analyze charging infrastructure.
- Calculate infrastructure pressure.

- Measure charging availability.
- Analyze highway accessibility.
- Identify potential charging-desert segments.
- Analyze commercial location opportunities.
- Evaluate proximity to power infrastructure.
- Develop an explainable, formula-driven scoring framework.
- Provide interactive exploration through Streamlit.
- Produce decision-oriented recommendations.

Stakeholders and Users

Stakeholder	Key Question
Government	Where should infrastructure support be prioritized?
Charging Operators	Where are potential expansion opportunities?
Investors	Which markets deserve further evaluation?
Urban Planners	Where are accessibility gaps?
Transport Planners	Which corridors need attention?
Analysts	What factors explain infrastructure pressure?

Product Vision

ChargeDesert aims to be a data-driven EV infrastructure intelligence platform that transforms fragmented mobility, geographic and infrastructure data into explainable insights for infrastructure planning and investment screening. The platform follows a consistent analytical pipeline:

**DATA → INFORMATION → ANALYSIS → GAP IDENTIFICATION
→ PRIORITIZATION → DECISION SUPPORT**

Project Scope

Included

- EV demand analysis
- Charging infrastructure analysis
- State analysis
- District analysis
- Highway analysis
- Spatial accessibility analysis
- POI analysis
- Power infrastructure analysis
- Infrastructure-gap indicators
- Opportunity scoring
- Interactive visualization
- Decision support

Not Currently Included as Validated Production Capabilities

- Real-time charger utilization
- Real-time traffic
- Actual land availability
- Electricity tariffs (beyond the illustrative cost_per_kwh field)
- Transformer capacity validation
- Land ownership
- Permitting requirements
- Commercial lease costs
- Actual investment returns

These limitations are discussed in full in Chapter 40.

Data Sources and Data Architecture

The analytical foundation was built from five primary data domains, supplied as sheets within DATA.xlsx:

Source Sheet	Records	Grain
EV_Registration	35,000	State × District × Vehicle Category × Year × Month
EV_Charging_Stations	15,000	Individual station
Highway_Corridors	10,000	Highway × Road segment
Points_Of_Interest	60,000	Individual POI
Power_Sub_Stations	10,000	Individual substation

These five raw source sheets form the foundation from which every analytical layer in this report — state, district, highway and location-opportunity — was derived.

Conceptual Data Architecture

EV REGISTRATIONS → DEMAND ANALYSIS CHARGING STATIONS + HIGHWAY DATA → GAP ANALYSIS POIs + SUBSTATIONS + ACCESSIBILITY → DECISION MODEL → RECOMMENDATIONS

Data Understanding

The project uses different analytical grains for different questions.

Analytical Layer	Grain	Records
State Performance	State	6
District Performance	State + District	30
Corridor Performance	Highway	5
Location Opportunity	State + District	30

The current analytical layer contains 6 state records, 30 district records, 5 highway records and 30 location-opportunity records. This distinction matters because different datasets should not automatically be joined simply because they appear related.

The Highway_Corridors sheet does not contain a state/district key (it is indexed by highway_id, segment_id, latitude and longitude only) and is therefore kept independent of the state/district analytical tables throughout this report, rather than joined to them without an appropriate geographic spatial join.

Data Engineering and Preparation

Raw Data → Data Profiling → Schema Validation → Missing Value Analysis → Duplicate Detection → Data Type Standardization → Geographic Validation → Feature Engineering → Dataset Integration → Analysis-Ready Data

Missing Values

Every field across all five source sheets was checked for missing observations. The dataset

contains zero missing values across all 1,30,000 records and every field, so no imputation or exclusion was required.

Duplicate Detection

Primary-key uniqueness was checked before aggregation: `record_id` (EV_Registration), `station_id` (EV_Charging_Stations), `poi_id` (Points_Of_Interest) and `substation_id` (Power_Sub_Stations) are all 100% unique with zero duplicates. In Highway_Corridors, the `segment_id` field is a per-highway sequence number (1–2,000) that repeats across the five highways by design; the true unique key, `highway_id` (e.g. "NH-48_SEG_0001"), has zero duplicates.

Geographic Validation

Latitude and longitude for every geo-tagged record (charging stations, highway segments, POIs, substations) were checked against a plausible bounding box for India (approximately 6°N–38°N, 68°E–98°E). No out-of-range coordinates were found in any of the four geo-tagged sheets.

Numeric Validation

Core count and volume fields (EV registrations, daily traffic volume, nearest-charger distance) were checked for negative or otherwise implausible values; none were found. Percentages and derived ratios computed in this report (availability %, EV penetration %, priority scores) are checked to fall within their expected 0–100 range as part of the analysis pipeline.

Analytical Methodology

The project distinguishes three types of metrics, each carrying a different degree of certainty.

Observed / Source Metrics

Directly recorded fields: EV registrations, station attributes (connectors, kW power, charger type, status), POI attributes (footfall index, parking capacity), and substation attributes (voltage, spare capacity, grid reliability). These are source-level observations, not calculations.

Derived Metrics

Arithmetic combinations of observed fields: EV penetration (%), availability rate (%), EV per operational charger, and connectors per 1,000 EVs. These are deterministic — the

same input data always produces the same derived value — but their business interpretation still requires the caveats discussed later.

Model Outputs

Demand score, supply-deficit score, infrastructure-readiness-gap score, commercial-activity score, gap score, priority score, priority class, corridor priority score, and every narrative recommendation built on them. These depend on the min–max normalization and weighting scheme defined later, and are explicitly treated throughout this report as illustrative model outputs based on unvalidated weights — not as ground truth.

KPI Framework

Total EV Registrations

Total EV registrations represented in the 2025 reference-year analytical population: 1,32,64,298. Higher registrations indicate greater observed EV demand.

Operational Charging Stations

Stations currently classified as "Operational" in the dataset: 11,178 of 15,000 total installed stations (74.5%).

EV per Operational Charger

$EV / Charger = EV \text{ Registrations} \div \text{Operational Charging Stations}$.

National value for 2025: 1,186.64. This acts as an infrastructure-pressure indicator — a higher value means more EV registrations are associated with each operational charging station. It should not be interpreted as actual charger utilization .

Charging Availability

$Availability = \text{Operational Stations} \div \text{Total Stations} \times 100$.

National value: 74.5%. Higher availability indicates a larger share of installed stations classified as operational and therefore usable.

High-Priority Locations

Counts locations classified into the High or Critical categories under the current priority model. The current project run contains 6 High-priority locations and 0 Critical-priority locations, for a combined High-Priority Locations KPI of 6 .

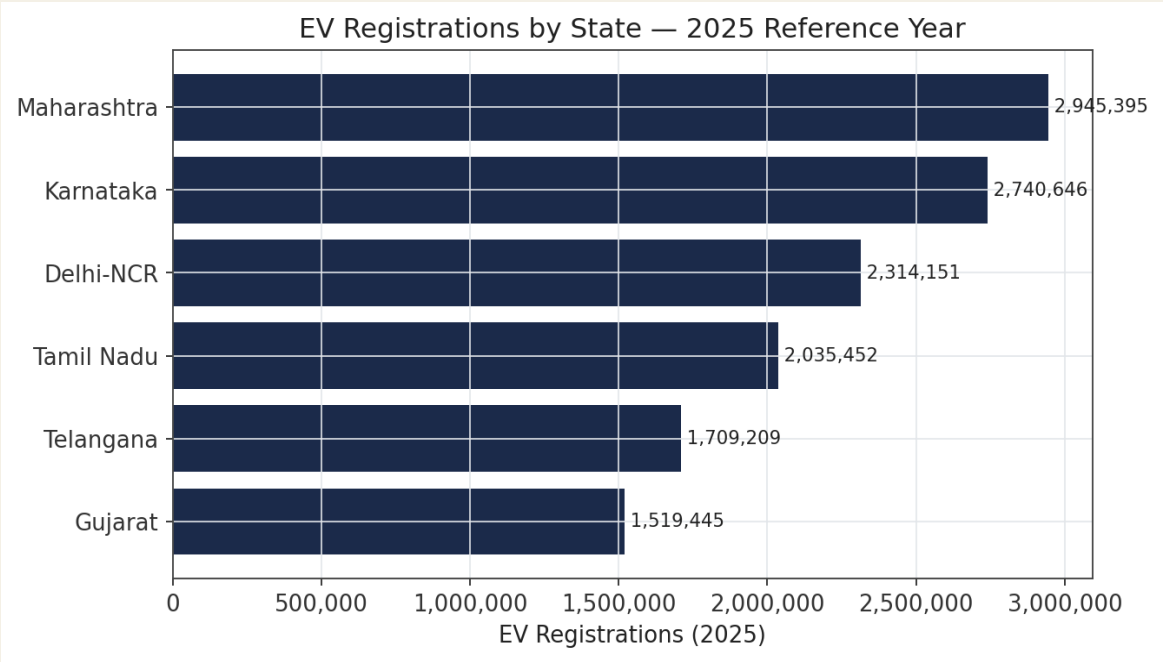
Exploratory Data Analysis

EDA was structured around five dimensions: Demand (EV registrations, penetration, temporal variation, category composition), Supply (total/operational stations, capacity, charger mix), Accessibility (nearest-charger distance, highway proximity, charging-desert segments), Opportunity (POI concentration, parking, commercial activity), and Readiness (proximity to power infrastructure).

EV Demand Analysis

EV registrations represent the primary demand signal. 1,32,64,298 EV registrations were recorded across 6 states and 30 districts in the 2025 reference year.

State Distribution



State	EV Registrations	Total Registrations	EV Penetration %
Maharashtra	29,45,395	1,78,94,283	16.5%
Karnataka	27,40,646	1,68,57,893	16.3%
Delhi-NCR	23,14,151	1,44,29,318	16.0%
Tamil Nadu	20,35,452	1,24,96,196	16.3%
Telangana	17,09,209	1,05,41,706	16.2%
Gujarat	15,19,445	96,61,744	15.7%

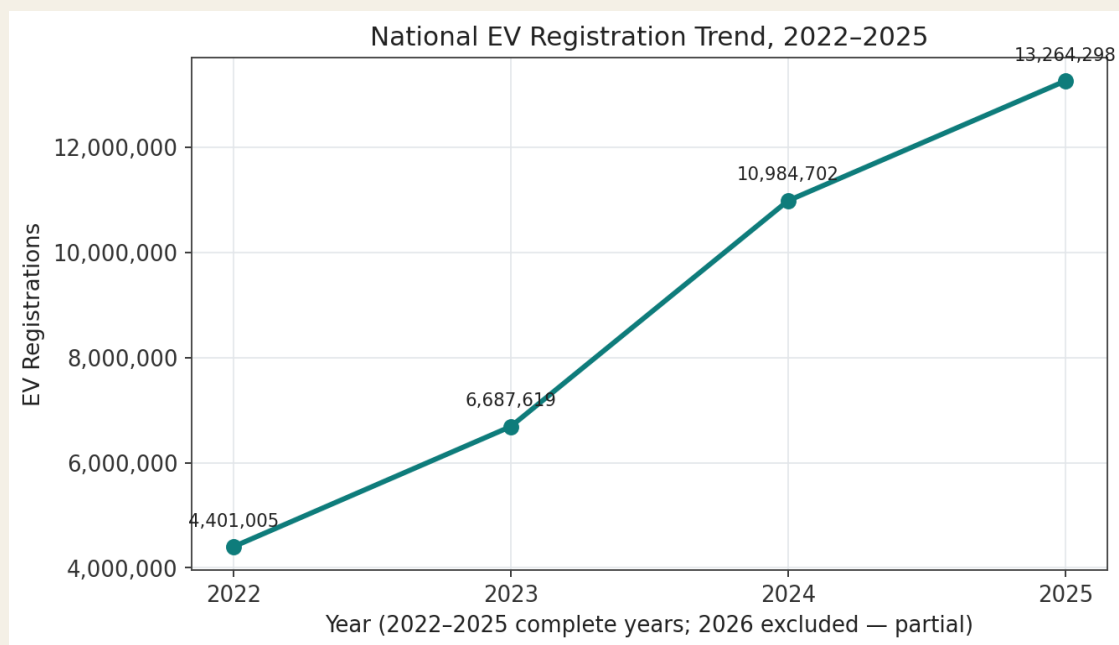
Maharashtra leads state-level EV registrations at 29,45,395 (16.5% penetration), followed by Karnataka (27,40,646).

District Distribution

The top 10 districts by 2025 EV registrations, out of 30 total districts, are shown below. This concentration indicates whether demand is spread evenly or concentrated in a small number of districts.

State	District	EV Registrations	Penetration %
Karnataka	Bengaluru Urban	12,14,343	16.0%
Maharashtra	Mumbai	10,28,860	16.8%
Maharashtra	Pune	8,93,137	16.8%
Tamil Nadu	Chennai	8,28,665	16.2%
Telangana	Hyderabad	7,69,228	15.8%
Delhi-NCR	South Delhi	5,82,036	16.5%
Delhi-NCR	Central Delhi	5,46,730	16.1%
Delhi-NCR	Gurugram	4,87,396	16.0%
Gujarat	Ahmedabad	4,74,002	16.2%
Karnataka	Bengaluru Rural	4,27,486	16.3%

Temporal Evolution

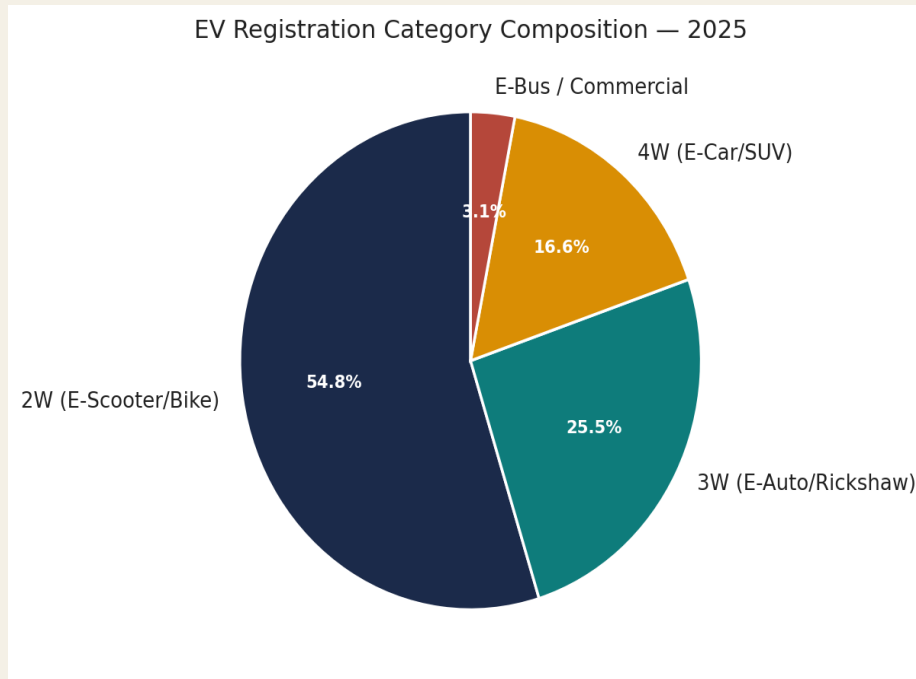


Years present in the source data: 2022, 2023, 2024, 2025, 2026. 2026 contains only the months Apr, Aug, Dec, Feb, Jan, Jul, Jun, Mar, May, Nov, Oct, Sep and is therefore excluded from year-over-year growth comparisons to avoid distortion.

Year	EV Registrations	YoY Growth %
2022	44,01,005	—
2023	66,87,619	52.0%
2024	1,09,84,702	64.3%

Year	EV Registrations	YoY Growth %
2025	1,32,64,298	20.8%

Category Composition

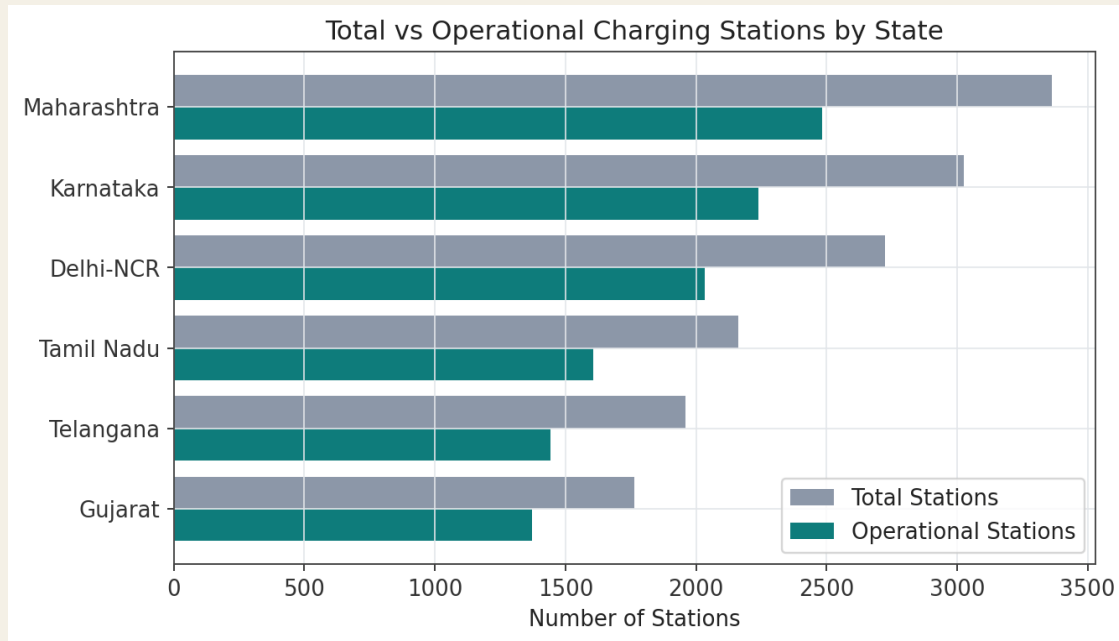


Vehicle Category	EV Registrations	Share %
2W (E-Scooter/Bike)	72,69,150	54.8%
3W (E-Auto/Rickshaw)	33,81,487	25.5%
4W (E-Car/SUV)	22,04,198	16.6%
E-Bus / Commercial	4,09,463	3.1%

The 2W (E-Scooter/Bike) category accounts for the largest share of 2025 registrations at 54.8%. Different vehicle categories may imply different mobility patterns and, by extension, different charging requirements; this report does not make unsupported claims about exact charger demand by category.

Charging Infrastructure Analysis

Charging infrastructure is evaluated using more than total station count: total stations, operational stations, availability, charging capacity, charger types, and geographic distribution.

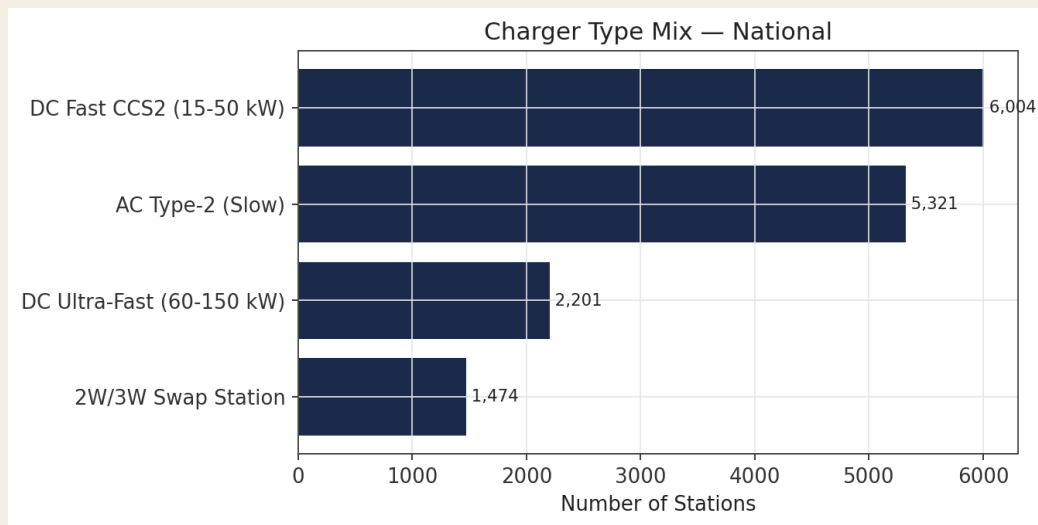


State	Total Stations	Operational	Availability %	24×7 %	Avg kW
Maharashtra	3,362	2,484	73.9%	80.6%	39.7
Karnataka	3,025	2,240	74.1%	79.0%	39.7
Delhi-NCR	2,725	2,032	74.6%	80.1%	39.5
Tamil Nadu	2,162	1,607	74.3%	80.7%	39.0
Telangana	1,961	1,443	73.6%	79.0%	39.3
Gujarat	1,765	1,372	77.7%	79.0%	40.1

Charger Status Breakdown (National)

Status	Count	Share %
Operational	11,178	74.5%
Occupied	2,304	15.4%
Under Maintenance	1,022	6.8%
Planned	496	3.3%

Charger Type Mix (National)



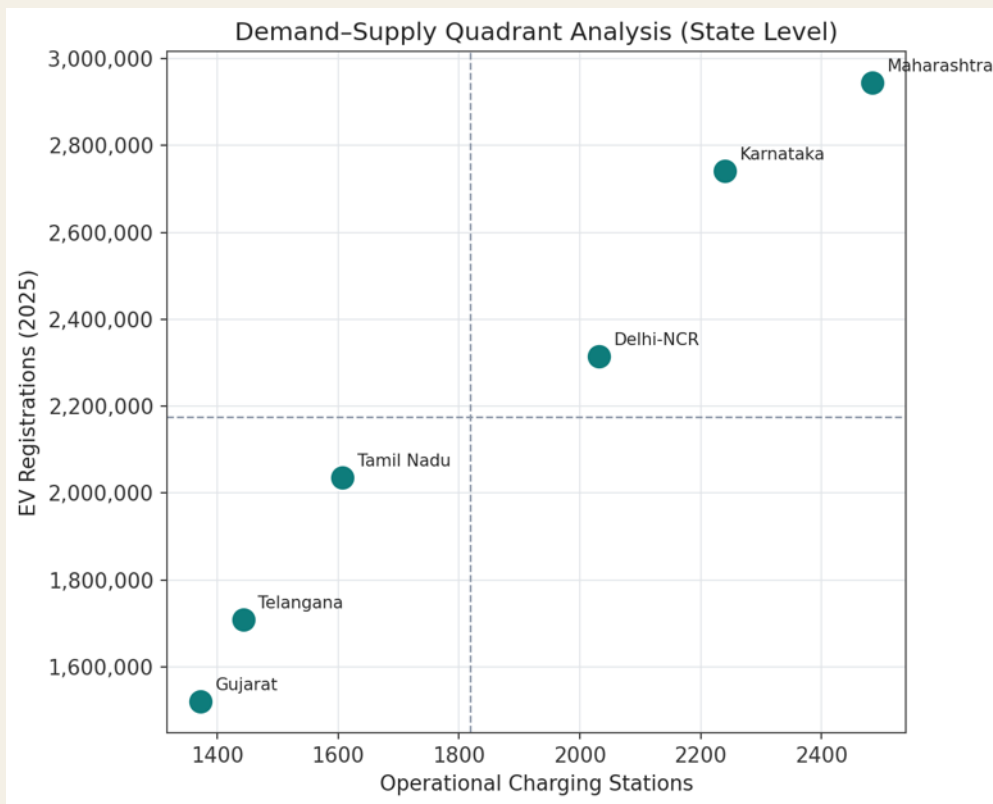
Charger Type	Count	Share %
DC Fast CCS2 (15-50 kW)	6,004	40.0%
AC Type-2 (Slow)	5,321	35.5%
DC Ultra-Fast (60-150 kW)	2,201	14.7%
2W/3W Swap Station	1,474	9.8%

The distinction between total and operational stations is important because infrastructure availability depends on operational status, not simply installed count — 1 in 4 installed stations nationally is currently non-operational (Occupied, Under Maintenance, or Planned).

Demand–Supply Gap Analysis

The central analytical concept is the relationship between EV demand and charging supply, expressed as $\text{Pressure} = \text{EV Registrations} \div \text{Operational Charging Stations}$. This allows regions to be compared using a common ratio.

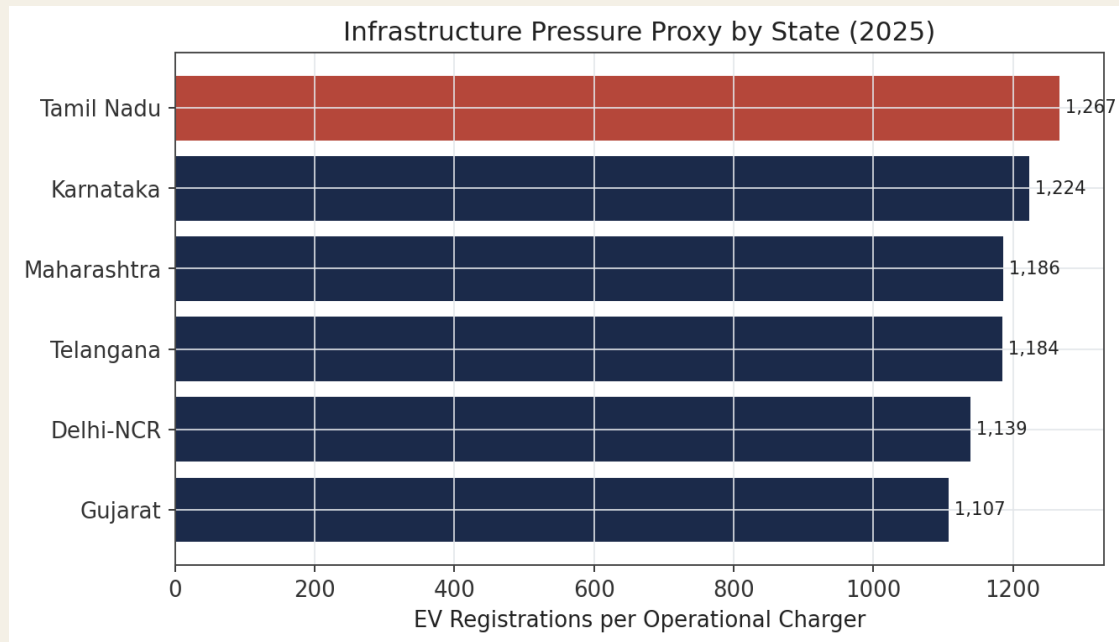
Demand–Supply Quadrants



State	EV Registrations	Operational Stations	Quadrant
Tamil Nadu	20,35,452	1,607	Low Demand / Low Supply
Karnataka	27,40,646	2,240	High Demand / High Supply
Maharashtra	29,45,395	2,484	High Demand / High Supply
Telangana	17,09,209	1,443	Low Demand / Low Supply
Delhi-NCR	23,14,151	2,032	High Demand / High Supply
Gujarat	15,19,445	1,372	Low Demand / Low Supply

States in the High-Demand / Low-Supply quadrant represent the strongest candidates for expansion investigation; High-Demand / High-Supply states are relatively mature markets that may still require capacity monitoring; Low-Demand states in either supply condition warrant ongoing monitoring rather than immediate prioritization. This framework is more informative than ranking states by EV registrations alone.

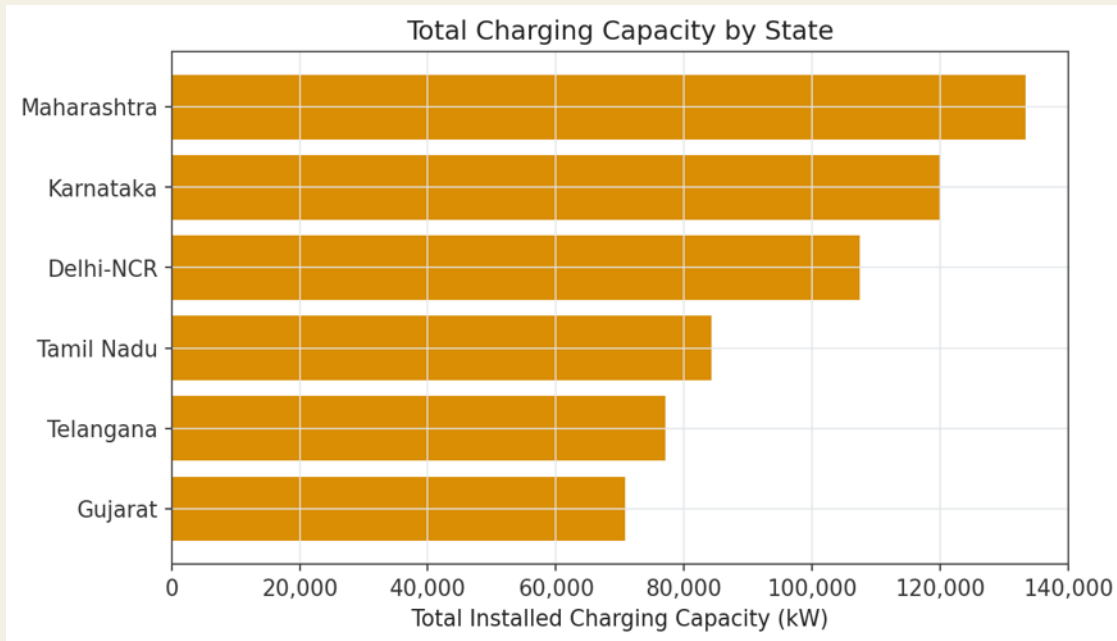
Charging Infrastructure Pressure



State	EV Registrations	Operational Stations	EV / Operational Charger
Tamil Nadu	20,35,452	1,607	1,266.6
Karnataka	27,40,646	2,240	1,223.5
Maharashtra	29,45,395	2,484	1,185.8
Telangana	17,09,209	1,443	1,184.5
Delhi-NCR	23,14,151	2,032	1,138.9
Gujarat	15,19,445	1,372	1,107.5

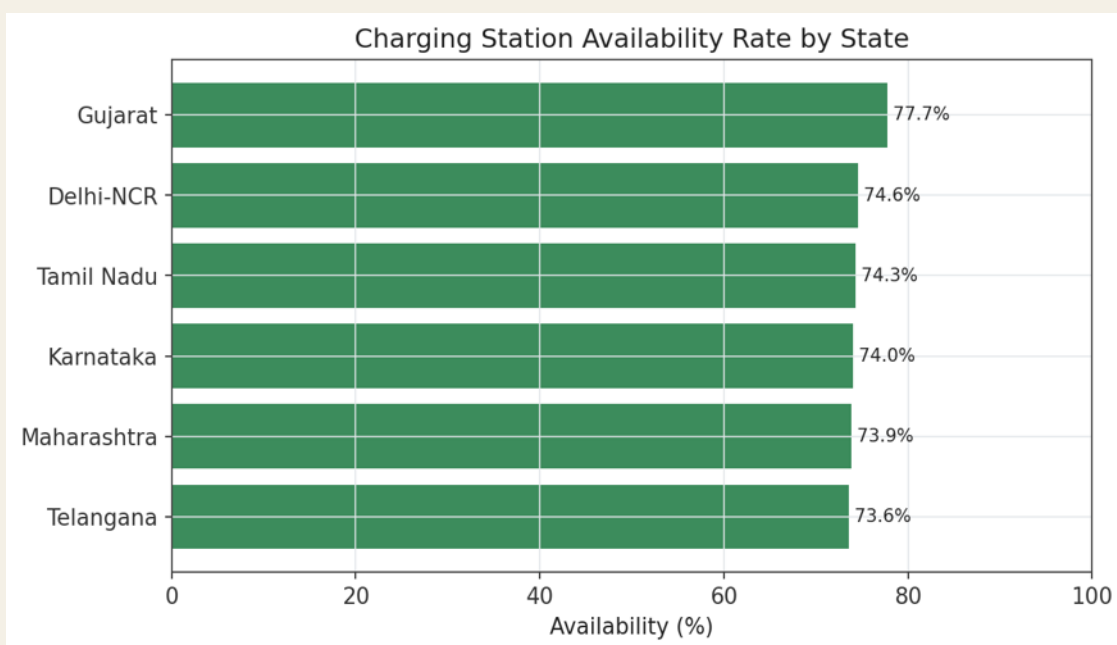
A higher ratio means more EV registrations relative to operational charging stations; a lower ratio means relatively greater charging availability per registered EV. This metric does not measure actual charger utilization, waiting time, charging duration, electricity demand, or station throughput — it is an Infrastructure Pressure Proxy, not a utilization metric.

Charging Capacity Analysis



State	Total Capacity (kW)	Avg kW / Station	Total Connectors
Maharashtra	1,33,349	39.7	14,286
Karnataka	1,19,964	39.7	13,110
Delhi-NCR	1,07,508	39.5	11,752
Tamil Nadu	84,236	39.0	9,288
Telangana	77,116	39.3	8,514
Gujarat	70,818	40.1	7,524

Charger Availability and Reliability



State	Total Stations	Operational	Availability %	24×7 %	Avg Cost/kWh (₹)
Maharashtra	3,362	2,484	73.9%	80.6%	19.07
Karnataka	3,025	2,240	74.1%	79.0%	19.05
Delhi-NCR	2,725	2,032	74.6%	80.1%	19.03
Tamil Nadu	2,162	1,607	74.3%	80.7%	19.03
Telangana	1,961	1,443	73.6%	79.0%	19.02
Gujarat	1,765	1,372	77.7%	79.0%	19.09

This table distinguishes operational stations (currently usable infrastructure) from installed stations, and reports 24×7 availability and average per-kWh cost as additional accessibility indicators, preventing the analysis from treating every installed station as equally usable.

Vehicle Category Analysis

EV registrations by vehicle category and state, 2025 reference year:

State	2W (E-Scooter/Bike)	3W (E-Auto/Rickshaw)	4W (E-Car/SUV)	E-Bus / Commercial
Delhi-NCR	12,99,985	5,56,168	3,74,390	83,608
Gujarat	8,14,049	4,07,026	2,49,421	48,949
Karnataka	14,85,115	7,46,437	4,30,804	78,290
Maharashtra	16,05,199	7,16,109	5,50,634	73,453
Tamil Nadu	11,43,923	4,97,200	3,23,583	70,746
Telangana	9,20,879	4,58,547	2,75,366	54,417

Business interpretation: different vehicle categories may imply different mobility patterns and therefore potentially different charging requirements (e.g. 2W/3W categories may rely more on distributed low-power charging, while 4W and E-Bus categories may concentrate demand at fewer, higher-capacity sites). This report avoids making unsupported claims about exact charger demand by vehicle category beyond what the registration and station-type data directly supports.

Geographic and Spatial Analysis

Spatial analysis is one of the project's strongest differentiators. The analysis considers charging-station distribution, proximity, accessibility, highway corridors, commercial locations, and power infrastructure.

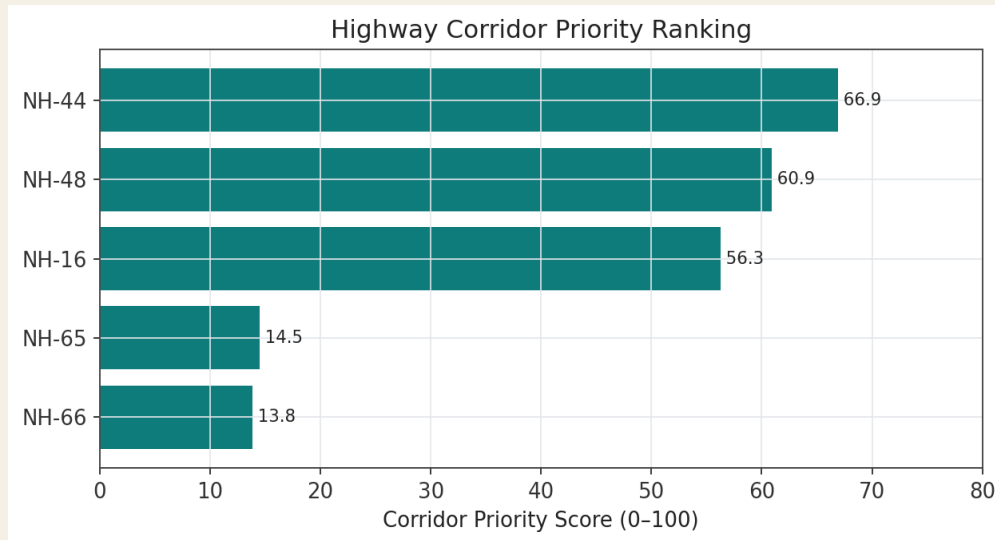
Charging Stations + EV Demand + Road Network + Administrative Boundaries → Spatial Analysis → Accessibility / Gap Indicators

All station, POI and substation records carry validated latitude/longitude coordinates, enabling district-level aggregation of infrastructure, commercial and grid-readiness

indicators used.

Highway Corridor Analysis

The corridor dataset contains 10,000 road-segment records across 5 highways. Of these segments, 3,493 (34.9%) are flagged as charging-desert segments.

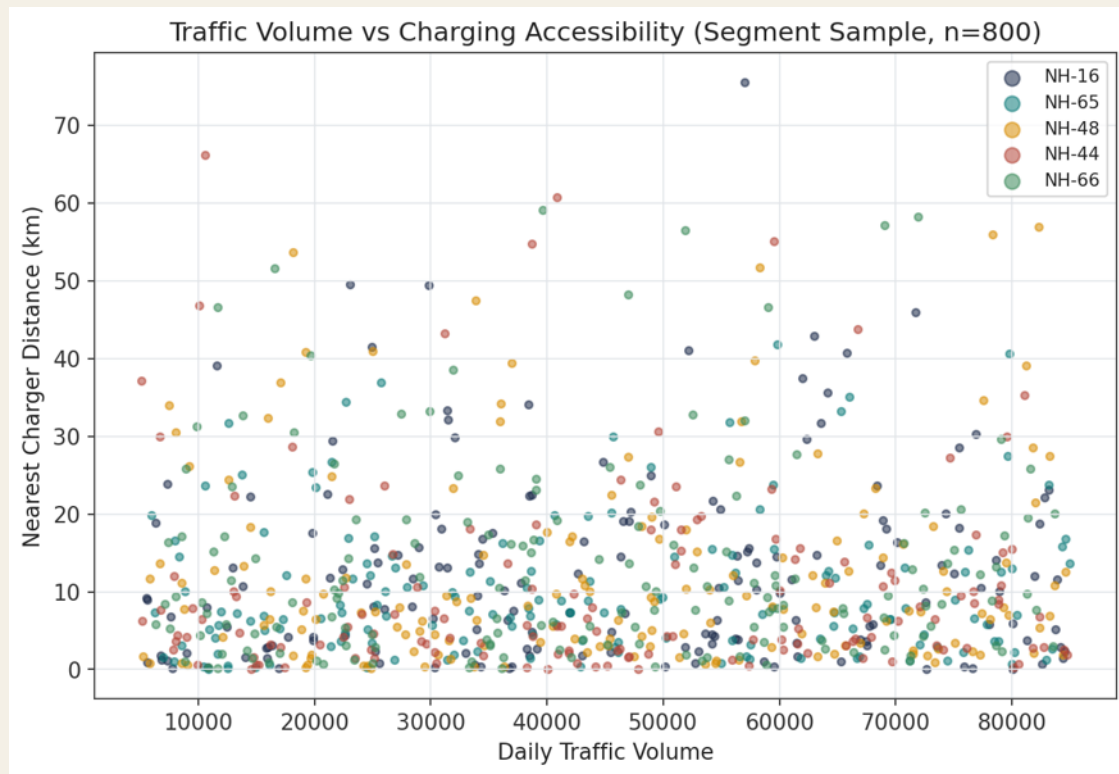


Rank	Highway	Avg Daily Traffic	Freight %	Desert Segment %	Priority Score
1	NH-44	45,257	30.2%	37.0%	66.9
2	NH-48	45,697	30.0%	34.9%	60.9
3	NH-16	45,316	29.9%	34.2%	56.3
4	NH-65	44,398	30.0%	34.5%	14.5
5	NH-66	44,603	29.7%	34.2%	13.8

NH-44 has the highest current model priority score (66.9), while NH-66 has the lowest (13.8). Because the score is model-derived, this ranking should be interpreted as a prioritization produced by the current analytical framework, not as an objective statement that any single corridor requires investment.

Traffic–Charging Accessibility Analysis

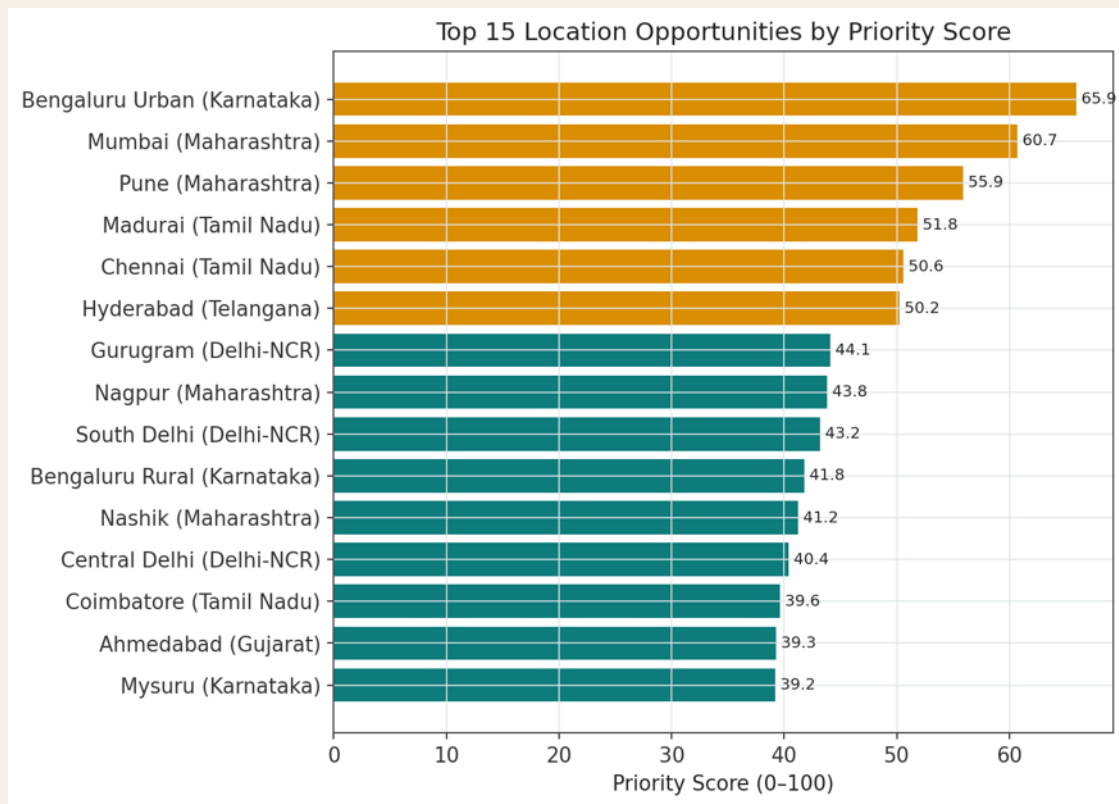
A useful analysis compares traffic volume against nearest-charging distance. The analytical question is: do high-traffic corridors also experience greater charging accessibility gaps?



Across all 10,000 segments, 2,512 segments (25.1%) fall in the High-Traffic / High-Charger-Distance quadrant — segments with above-median traffic and above-median distance to the nearest charger. These segments may represent potentially important infrastructure opportunities; however, this analysis does not automatically classify them as investment-ready sites without further site-level due diligence.

Location Opportunity Analysis

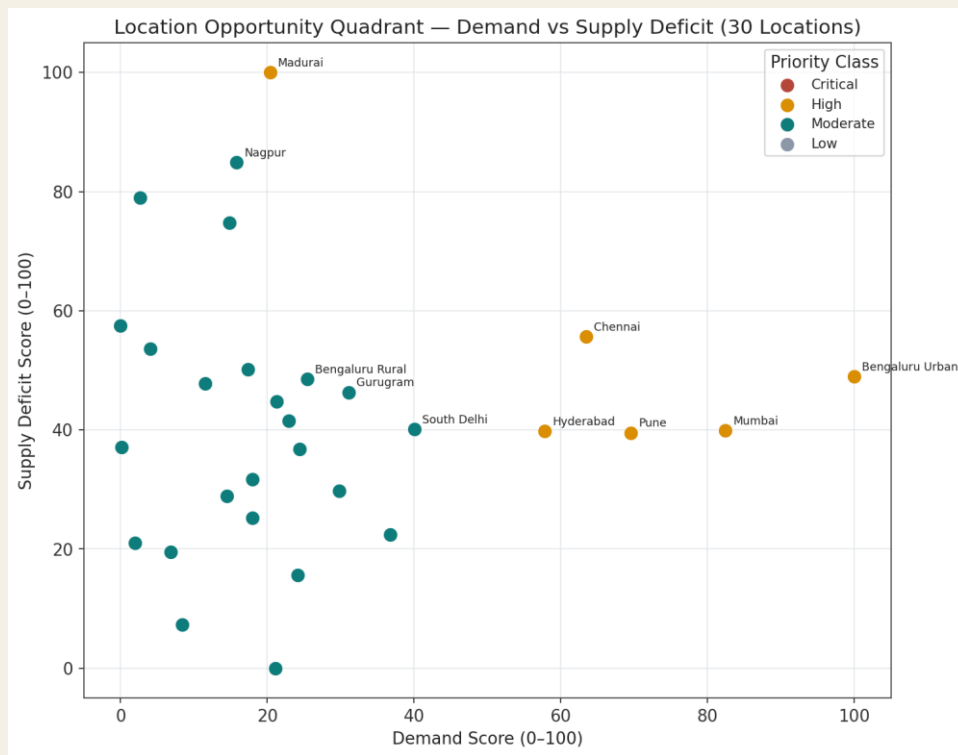
Location opportunity analysis combines demand, supply deficit, infrastructure readiness and commercial activity indicators across 30 district-level location-opportunity records.



Top 10 Ranked Locations

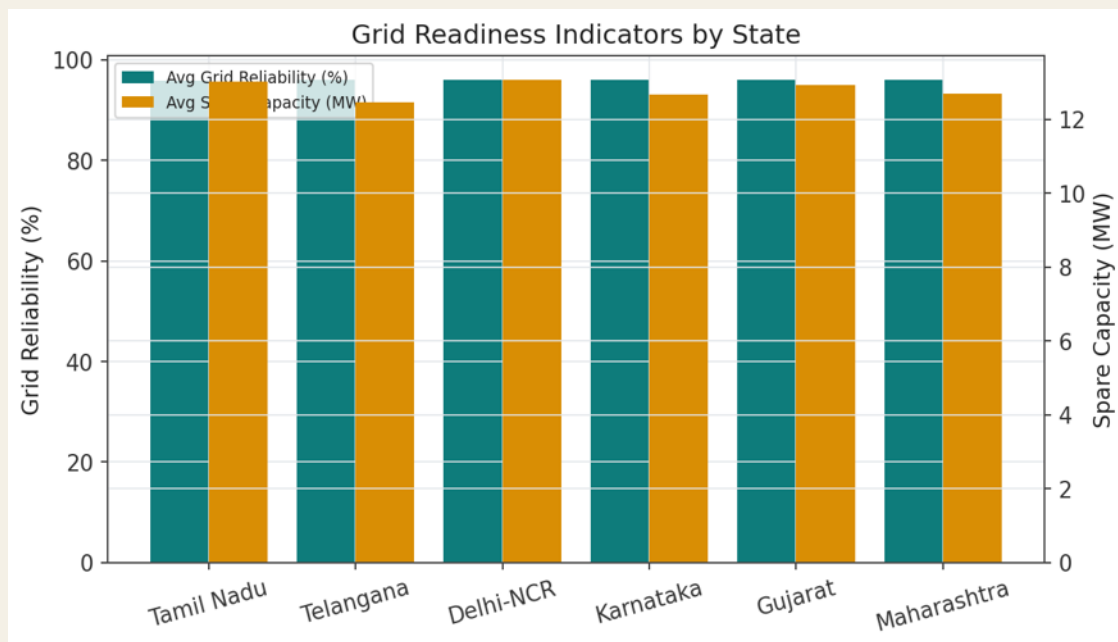
Rank	State	District	Priority Score	Class
1	Karnataka	Bengaluru Urban	65.9	High
2	Maharashtra	Mumbai	60.7	High
3	Maharashtra	Pune	55.9	High
4	Tamil Nadu	Madurai	51.8	High
5	Tamil Nadu	Chennai	50.6	High
6	Telangana	Hyderabad	50.2	High
7	Delhi-NCR	Gurugram	44.1	Moderate
8	Maharashtra	Nagpur	43.8	Moderate
9	Delhi-NCR	South Delhi	43.2	Moderate
10	Karnataka	Bengaluru Rural	41.8	Moderate

These rankings are explicitly model outputs and should be presented as illustrative candidate opportunities for further investigation, not as validated investment recommendations.



Grid Readiness Analysis

Charging infrastructure requires access to electrical infrastructure. Location suitability therefore considers proximity to power infrastructure where data is available: number of substations, spare capacity, voltage class, and grid reliability.



Proximity to a substation does not prove that sufficient electrical capacity is available for a new high-power charging deployment. Grid proximity is therefore treated throughout this report as a readiness indicator, not confirmed grid capacity.

Multi-Criteria Decision Analysis

The project uses a weighted decision framework that evaluates multiple business considerations simultaneously:

Demand + Supply Deficit + Infrastructure Readiness Gap + Commercial Activity → GAP SCORE → (combined with) Infrastructure Readiness → PRIORITY SCORE

Priority Scoring Framework

The current project methodology defines, at the district level:

GapScore = $0.35 \times \text{Demand Score} + 0.30 \times \text{Supply-Deficit Score} + 0.20 \times \text{Infrastructure-Readiness-Gap Score} + 0.15 \times \text{Commercial-Activity Score}$

PriorityScore = $0.70 \times \text{GapScore} + 0.30 \times \text{Infrastructure Readiness Score}$

Each component score is computed by min–max normalizing the underlying district-level metric to a 0–100 scale: Demand Score from 2025 EV registrations; Supply-Deficit Score from the EV-per-operational-charger ratio; Infrastructure Readiness Score from the average of grid reliability %, substation spare capacity, and station availability % (with the Readiness-Gap Score used inside GapScore being its complement, 100 minus readiness); and Commercial-Activity Score from the average of normalized POI footfall, parking capacity and POI count. This normalization approach and the weights above are explicitly illustrative.

The priority classification thresholds are:

Gap Score	Priority
≥ 75	Critical
≥ 50	High
≥ 25	Moderate
< 25	Low

Current Distribution

Class	Current Count
Critical	0
High	6
Moderate	24
Low	0

The current High-Priority Locations KPI of 6 is composed of 6 High-classified districts and 0 Critical-classified districts.

Corridor Priority Model

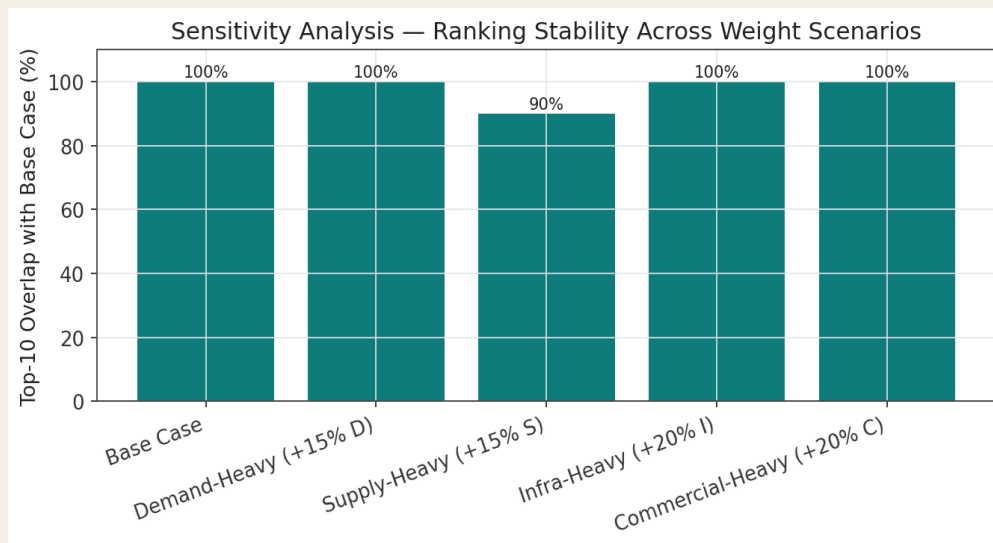
The corridor priority score is defined as:

$$\text{CorridorPriority} = 0.35 \times \text{Traffic (normalized)} + 0.20 \times \text{Freight \% (normalized)} + 0.25 \times \text{Nearest-Charger-Distance (normalized)} + 0.20 \times \text{Desert-Segment \% (normalized)}$$

The components are min–max normalized across the five highways before the weighted score is calculated. This means the model gives greatest weight to traffic volume, followed by nearest-charger distance, desert-segment percentage, then freight percentage. As with the location priority model, this should be read as an illustrative prioritization framework.

Sensitivity Analysis

A robust decision model should not depend entirely on one arbitrary set of weights. Five scenarios were tested: a Base Case (documented weights), and four perturbed cases that increase one GapScore component's weight by 15–20% while proportionally reducing the others, renormalized to sum to 1.0.



Scenario	Top-10 Overlap with Base Case	Overlap %
Base Case	10 / 10	100.0%
Demand-Heavy (+15% D)	10 / 10	100.0%
Supply-Heavy (+15% S)	9 / 10	90.0%
Infra-Heavy (+20% I)	10 / 10	100.0%
Commercial-Heavy (+20% C)	10 / 10	100.0%

The top-10 priority district list remains substantially stable across all four perturbed weighting scenarios, indicating that the current ranking is not driven by a single fragile assumption. This mirrors the $\pm 10\text{--}20\%$ weight-variation approach recommended in the project methodology.

Recommendation Stability

A recommendation is stronger if it remains near the top across reasonable changes in assumptions. The table below shows each base-case top-10 district's rank range across the five scenarios, and a stability label (High: range ≤ 2 ranks; Medium: range ≤ 5 ranks; Low: range > 5 ranks).

District	Base Rank	Rank Range Across Scenarios	Stability
Bengaluru Urban	1	0	High
Mumbai	2	0	High
Pune	3	0	High
Madurai	4	1	High
Chennai	5	1	High
Hyderabad	6	0	High
Gurugram	7	1	High
Nagpur	8	2	High
South Delhi	9	1	High
Bengaluru Rural	10	1	High

This distinguishes "the model says this district is #1" from "the recommendation remains strong under reasonable assumption changes" — the latter is the more decision-useful statement.

Scenario Analysis

Scenario analysis simulates the effect of increasing operational charging supply nationally, recalculating $\text{EV} / \text{Charger} = \text{EV Registrations} \div \text{Projected Operational Stations}$ for each scenario. These are scenario simulations, not forecasts.

Operational Stations Increase	Projected Operational Stations	EV per Charger
0%	11,178	1,186.6
10%	12,296	1,078.8
25%	13,972	949.3
50%	16,767	791.1

A 50% increase in operational stations would reduce the national infrastructure-pressure proxy from 1,186.6 to 791.1 EV registrations per operational charger — illustrating how

the pressure proxy responds to supply expansion, holding 2025 demand constant.

Key Findings

Finding 1 — EV Demand and Charging Supply Should Be Evaluated Together

A high EV-registration state does not automatically mean it is underserved. Maharashtra leads in absolute EV registrations (29,45,395), but the highest infrastructure-pressure state by EV-per-operational-charger is Tamil Nadu (1,267). Infrastructure pressure must be evaluated relative to operational charging supply, not registrations alone.

Finding 2 — Operational Availability Matters

Nationally, only 74.5% of the 15,000 installed stations are currently Operational. Installed stations should not be treated as equivalent to usable stations; operational availability provides a more meaningful infrastructure indicator than raw station count.

Finding 3 — Highway Analysis Reveals a Different Dimension of Infrastructure Need

The corridor analysis indicates real variation in the current model-derived priority scores: NH-44 ranks highest (66.9) and NH-66 ranks lowest (13.8), driven primarily by differences in traffic volume and nearest-charger distance.

Finding 4 — Location Prioritization Is Multi-Dimensional

The current model's top-ranked opportunity is Bengaluru Urban (Karnataka), followed by Mumbai and Pune among the leading locations. However, as with all model outputs in this report, these rankings should be interpreted as candidates for further investigation rather than validated investment recommendations.

Finding 5 — A Single Metric Is Insufficient

EV registrations, charging stations, accessibility, commercial activity and infrastructure readiness each capture a different dimension of the problem. The sensitivity analysis in Section 31 shows the multi-criteria ranking is reasonably stable under weight perturbation, supporting the case for a multi-criteria framework over any single ranking metric.

Business Recommendations

Recommendation 1 — Prioritize Investigation of High-Demand / High-Pressure Areas

States and districts with high EV demand and comparatively limited operational charging infrastructure — led by Bengaluru Urban, Mumbai and Pune in the current model — should be investigated first.

Recommendation 2 — Evaluate Highway Corridors Separately

Highway planning should incorporate traffic, freight percentage, nearest-charging distance and desert-segment percentage, with NH-44 as the current highest-priority corridor for further investigation.

Recommendation 3 — Consider Commercial Opportunity

Potential charging locations should be evaluated alongside parking capacity, commercial activity, destination footfall and accessibility, as captured in the Commercial Activity Score component of the priority model.

Recommendation 4 — Incorporate Grid-Readiness Information

Potential sites should be evaluated against electrical infrastructure — substation proximity, spare capacity and grid reliability — before investment decisions are made.

Recommendation 5 — Use Scenario and Sensitivity Analysis Before Deployment

Decision-makers should avoid treating any single model ranking as final; rankings and pressure indicators shift under alternative assumptions.

Stakeholder-Specific Recommendations

Stakeholder	Recommended Use of the Platform
Government	Use the platform as an analytical screening tool for identifying areas requiring further investigation.
Charging Operators	Use demand, accessibility and opportunity indicators to generate candidate expansion locations.
Investors	Use the model to create a preliminary opportunity shortlist, followed by commercial and technical due diligence.
Urban Planners	Use district-level and accessibility analysis to identify underserved areas.
Highway Authorities	Use corridor analysis to investigate high-traffic segments with potentially poor charging accessibility.

Streamlit Application

The Streamlit application is the interactive implementation of the analytical framework described in this report. Its role is complementary to the report:

Component	Role
Streamlit Application	Interactive exploration, filtering, maps, KPI visualization, opportunity analysis, scenario exploration, recommendation views.
This Report	Methodology, analytical reasoning, detailed calculations, deeper analysis, validation, limitations, recommendations.

In short: Streamlit = Explore; Report = Understand. Screenshots of the live application are maintained separately from this analytical document and are not reproduced here.

System Architecture

DATA SOURCES (EV Registrations, Charging Data, Highway Data, POIs, Substations) → DATA PROCESSING (Spatial Processing + Analytical Processing) → DECISION ENGINE → STREAMLIT APP → INTERACTIVE ANALYSIS + REPORT & PRESENTATION

Data Quality and Validation

Structural and Completeness Validation

Sheet	Rows	Missing Values	Duplicate Primary Keys	Geo Out-of-Bounds
EV_Registration	35,000	0	0	—
EV_Charging_Stations	15,000	0	0	0
Highway_Corridors	10,000	0	8000 (segment_id; 0 on highway_id)	0
Points_Of_Interest	60,000	0	0	0
Power_Sub_Stations	10,000	0	0	0

All five sheets pass structural and completeness validation with zero missing values across every field. The apparent segment_id duplication in Highway_Corridors is expected: segment_id is a 1–2,000 sequence number reused per highway, while the true unique key, highway_id, has zero duplicates.

Business-Rule Validation

- EV Registrations ≥ 0 — passed, 0 negative values.
- Charging station counts ≥ 0 — passed.
- Daily traffic volume ≥ 0 — passed, 0 negative values.
- Nearest-charger distance ≥ 0 — passed, 0 negative values.
- All computed percentages (availability, penetration, priority score) fall within their expected 0–100 range by construction (min–max normalization and ratio definitions).

Model Validation

Weight sensitivity and ranking stability were tested explicitly ; the top-10 priority district list remained substantially stable under all tested perturbations.

Limitations

Synthetic Data

The dataset underlying this report is simulated / synthetic portfolio data. Findings should not be interpreted as official statistics.

EV Registrations $\neq$ Active EV Population

Registrations indicate recorded registrations, not necessarily active vehicles currently operating in the region.

EV / Charger $\neq$ Utilization

A high EV-per-charger ratio does not necessarily mean chargers experience high utilization. Actual utilization would require charging-session data, energy delivered, session duration, occupancy and waiting time — none of which are present in the source dataset.

Distance $\neq$ Travel Accessibility

Geographic distance does not necessarily equal actual travel time. Road conditions, traffic, route availability and network topology can change real accessibility.

Grid Proximity $\neq$ Grid Capacity

A nearby substation with reported spare capacity does not guarantee that additional charging infrastructure can actually be connected without further electrical engineering review.

Commercial Activity $\neq$ Commercial Feasibility

Commercial potential (footfall, parking) requires additional factors not in this dataset: land cost, lease cost, expected utilization, operating cost, revenue, and permitting.

Model Weights Are Illustrative

The scoring framework uses weighted assumptions that have not been externally validated. These scores are explicitly illustrative model outputs.

Corridor Data Cannot Automatically Be Linked to District Data

The Highway_Corridors sheet lacks a shared state/district key. A proper geographic spatial join (segment coordinates against district boundaries) would be required for integrated corridor–district analysis; this report keeps the two analytical layers separate throughout.

Risks and Assumptions

Risk	Description
Data coverage	The dataset may not capture every infrastructure asset in the six covered states.
Temporal mismatch	Different sheets may implicitly represent slightly different observation periods.
Model sensitivity	Different weight assumptions can change rankings, shows reasonable stability for the top-10 list.
Geographic bias	The six covered states and thirty districts may not be representative of India as a whole.
Infrastructure status	Operational status may change over time and is only a point-in-time snapshot.

Conclusion

ChargeDesert demonstrates an end-to-end approach to EV infrastructure analytics. Rather than simply visualizing charging-station locations, the project connects EV demand, charging supply, accessibility, transportation, commercial opportunity and infrastructure readiness into a unified decision-support framework, computed directly from 1,30,000 underlying records across five data domains.

The project demonstrates how data analytics can move from Descriptive Analysis, to Diagnostic Analysis, and ultimately toward Decision Intelligence. The most important contribution of the platform is therefore not a single map, KPI or score — it is the analytical workflow: What is happening? → Where is it happening? → Why might the gap exist? → Which locations deserve attention? → How robust is the recommendation? → What should decision-makers investigate next?

The current version should be considered a portfolio and analytical demonstration, not a production investment-planning system. Future integration of validated real-world datasets, real-time infrastructure information, electrical capacity data, live traffic data, land economics and observed charging utilization would be required before operational deployment.